Manish Kudtarkar

Software Engineer | Full Stack, Data & Enterprise Systems

Mumbai, Maharashtra, India | +91 9324377433 | manishkudtarkar467@gmail.com | Portfolio

linkedin.com/in/manish-kudtarkar | github.com/ManishKudtarkar

Professional Summary

Computer Science undergraduate specializing in Big Data Analytics with hands-on experience in full-stack enterprise architecture, relational database engineering, REST API development, and data pipelines. Proven background building centralized campus platforms with granular Role-Based Access Control (RBAC), developing automated ETL frameworks, and containerizing microservices on AWS (ECS, RDS). Strong core foundation in object-oriented programming (OOP), relational schema modeling, and production debugging, well-prepared for modern enterprise systems.

Technical Skills

- **Programming Languages:** Java, Python, C, C++, JavaScript, TypeScript, SQL
- **Core Computer Science:** Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), DBMS, Operating Systems
- **Databases & ORM:** PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, Relational Schema Design, Query Optimization
- **Software Development:** Node.js, Express.js, React 18, Next.js, FastAPI, Flask, RESTful APIs, JWT Authentication, Microservices
- **DevOps & Cloud:** AWS (ECS, ECR, RDS, EC2, S3, IAM, Lambda, Amplify, CloudFront), Docker, Git, GitHub Actions (CI/CD), Linux
- **Data Engineering & Analytics:** ETL Pipelines, Schema Validation, Parquet, Apache Airflow, Apache Hive, Hadoop HDFS, Pandas

Professional Experience

Full Stack Developer — **January 2025 – July 2026**

Axflo Pump Solution — Mumbai, India (Remote)

- Collaborated on end-to-end software development lifecycle, translating enterprise business requirements into responsive full-stack solutions and modular relational data models.
- Engineered and unit-tested 8+ REST API endpoints integrated with persistent relational databases, enforcing schema validations and error handling across critical workflows.
- Automated business inquiry processing pipelines, resulting in an estimated 40% reduction in manual handling time and significantly improving operational record consistency.
- Maintained automated GitHub Actions CI/CD deployment pipelines, managing multi-stage production builds and troubleshooting runtime defects.

Selected Projects

PUVerse – Campus Management Platform | *Next.js, Express, PostgreSQL, Prisma, Docker, AWS* **Project Link**

- Architected a centralized full-stack campus management platform featuring dynamic administrative dashboards and fine-grained Role-Based Access Control (RBAC) across student and faculty tiers.
- Designed a normalized PostgreSQL schema with Prisma ORM, deploying containerized microservices on Amazon ECS and Amazon RDS with secure parameter configurations via AWS SSM.

EcoPredict – AI Biodiversity Forecasting System | *Python, Pandas, ML, GBIF, Scikit-learn* **Project Link**

- Developed an AI-driven biodiversity analysis system processing 2,868 raw GBIF insect occurrence records into a curated dataset of 2,633 valid entries to evaluate species distribution patterns.
- Built a custom Python ETL pipeline handling duplicate resolution, missing values, coordinate sanitization, and noisy observations to prepare model artifacts for forecasting tasks.

OpenIngest – Enterprise ETL Framework | *Python, Airflow, Hadoop HDFS, S3, Parquet* **Project Link**

- Designed and published an extensible Python ETL library on PyPI supporting multi-cloud storage ingestion (S3, Azure Blob, GCS) with optimized Parquet columnar serialization and schema validation.
- Automated generation of production-ready Apache Airflow Directed Acyclic Graphs (DAGs) for recurring enterprise data pipelines.

Industrial Smart Grid AI – Distributed ML Platform | *Python, FastAPI, Scikit-learn* **Project Link**

- Engineered a distributed telemetry platform delivering 25 FastAPI endpoints across a 42-feature pipeline, conducting automated anomaly forecasting and fault detection with sub-50ms latency.

Education

Parul University — **Expected May 2027**

Bachelor of Technology in Computer Science Engineering (Big Data Analytics) — Vadodara, Gujarat

G.R. Patil College — **June 2023**

Higher Secondary Certificate (HSC) – Science and Computer Science — Mumbai, Maharashtra

Certifications

- **AWS Certified Cloud Practitioner** – Amazon Web Services — Feb 2026 – Feb 2029
- **Data Analyst Associate** – DataCamp | **Data Engineering** – DataCamp